The Myrmidon Swarm

Insect-Scale Repair Robots with Shared Learning for Autonomous Spacecraft Maintenance

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Abstract

A spacecraft hull is under continuous, invisible attack. Micrometeoroids and orbital debris strike it constantly, thermal cycling works its joints, and radiation embrittles its surfaces — yet nobody is out there watching. Damage accumulates unseen between the rare, expensive, dangerous occasions when a human puts on a suit and goes outside. This study proposes replacing that model entirely with a permanent, insect-scale maintenance crew. The MY-1 'Myrmidon' swarm is a population of thousands of small robots that live on the outside of a spacecraft, crawling its surface indefinitely, finding micro-damage while it is still micro, and repairing it in place. Their decisive feature is not mechanical but cognitive: the swarm learns collectively. When one robot works out how to handle an unfamiliar kind of crack, that knowledge propagates to every other robot in the swarm and, in time, to swarms on other vehicles — one learns, all know. We describe the robots, the collective behaviour, the shared-learning architecture, and a staged path from inspection to autonomous repair. Every ingredient — swarm robotics, micro-robots, federated learning — exists today in laboratory form.

1. Introduction: Nobody Is Watching the Hull

The outside of a spacecraft is the part most exposed and least observed. It is struck continually by micrometeoroids and small orbital debris, whose hypervelocity impacts leave craters, spalls, and cracks that shielding is designed to survive but not to heal (Christiansen, 2003). It flexes through brutal thermal cycles every orbit, cold to hot and back, working at seams and adhesives. Radiation and atomic oxygen degrade its outer layers. And for the entire operating life of the vehicle, almost none of this is inspected, because inspection means either a robotic arm with a camera or a human being in a suit — both scarce, both expensive, both slow.

The result is a maintenance philosophy built on ignorance and overdesign: build the structure strong enough to tolerate damage nobody will find, and hope. This is tolerable for a mission of a few years in low orbit with resupply. It is untenable for the machines this catalogue imagines — stations that must last decades, transit vehicles that cannot turn back, habitats around other worlds where no rescue is coming. Those vehicles need what a termite mound has and a spacecraft does not: a permanent population of small workers who never stop tending the structure.

2. Concept Description

The MY-1 'Myrmidon' is a robot roughly the size of a large insect, and the swarm is thousands of them living permanently on the spacecraft's exterior. Each carries the minimum needed to be useful: a means of holding to and crawling across the hull, close-range sensors to inspect the surface it walks over, a small repair capability, a short-range radio to talk to its neighbours, and enough onboard processing to decide what to do next. They are individually simple, individually expendable, and collectively tireless — they patrol the hull continuously, so damage is found in the hours after it happens rather than in the years after, when it has grown.

Their organisation is drawn from social insects and from the swarm-robotics tradition it inspired, in which large numbers of simple units produce robust collective behaviour without any central controller (■ahin, 2005). No Myrmidon is in charge. Coverage of the hull emerges from local rules and local communication; when one robot finds damage it cannot handle alone, it recruits neighbours, and a work party assembles at the site and disperses when the job is done. The loss of any individual — and individuals will be lost — changes nothing about the swarm's ability to do its work, which is precisely the property a decades-long mission needs.

3. The Decisive Feature: One Learns, All Know

What separates the Myrmidon concept from a simple population of repair robots is that the swarm is a single learning system. Each robot encounters damage, attempts a repair, and observes whether it worked — a stream of hard-won, real-world experience. Rather than each robot hoarding its own lessons, the swarm shares them: what each unit learns from its local experience is aggregated into a common model of how to recognise and treat damage, and the improved model is distributed back to every unit. A crack type solved once, anywhere on the hull, is thereafter known everywhere on the hull.

This is federated learning, and it is not speculative: the technique of training a shared model from many devices' local experience, without pooling the raw data centrally, is well established and widely deployed on terrestrial device fleets (McMahan et al., 2017). Its properties suit a spacecraft swarm unusually well. Communication stays modest, since units exchange model updates rather than raw sensor streams — important when bandwidth is scarce. Learning continues without contact with Earth, which matters when light-lag makes ground supervision impossible. And the same mechanism extends across vehicles: a lesson learned by a swarm on a Mars transit ship can, on the next communications pass, become part of the model carried by swarms everywhere. The fleet's maintenance competence compounds over time rather than resetting with each new mission.

4. The Machine Itself

The individual robot's hardest problem is staying attached. In microgravity there is nothing to hold a walking robot down, so a Myrmidon must grip its hull deliberately — through dry adhesion of the kind that lets a gecko hold to glass, through magnetism where the structure permits, or through mechanical purchase on prepared surfaces — and must never let go accidentally, since a released robot becomes debris. Its locomotion must cross a landscape of panels, seams, handrails, and instruments without snagging. Its power comes from small solar cells, supplemented by docking at charging points, since a body this size has almost no room for stored energy.

Its repair capability is deliberately limited to the small: sealing a micro-crack, applying a patch material over a pit, cleaning a contaminated surface, restoring a coating, or flagging what it cannot fix for a larger robot or a human. This modesty is the point. The swarm's value is not that it can perform major surgery but that it performs continuous preventive care — catching the crack at a millimetre so that it never becomes a breach, which is exactly the maintenance that spacecraft currently cannot get. Thousands of such small interventions, applied constantly over decades, change the ageing curve of a structure more than any single dramatic repair.

5. Operations Concept

Phase one is inspection only, and is close to achievable today: a modest population of crawlers that traverse the exterior of a station, image everything they pass, and build a continuously-updated map of the hull's condition — valuable immediately, and free of any risk from a robot attempting an unwise repair. Phase two adds simple intervention on the easiest damage classes, with the shared-learning loop running so that the swarm's competence improves from real experience. Phase three is the mature system: a permanent, self-managing maintenance population on long-duration vehicles, autonomously finding and fixing the small damage of decades, its accumulated model shared across an entire fleet. Human spacewalks become reserved for what genuinely requires a human, which is the point.

Parameter	Phase 1 (Inspect)	Phase 3 (Autonomous care)
Swarm size	tens	thousands
Unit	insect-scale crawler	same, refined
Capability	image + map hull	find + repair micro-damage
Learning	data collection	federated, fleet-wide
Ground role	supervises	occasional model sync
Effect	damage becomes visible	damage rarely grows

Table 1. Representative parameters for the staged build-out.

6. Honest Difficulties

The obstacles are practical and several. Adhesion is the first and hardest: a robot that must cling reliably for years across temperature extremes, vacuum, and contamination — and that must never detach and become debris — is a genuine materials and mechanisms challenge. Power is the second, since an insect-scale body has minimal room for solar cells or batteries and must budget every action. Miniaturised, radiation-tolerant electronics are the third. And the collective-behaviour problem is real: coordinating thousands of units to cover a complex surface without collision, congestion, or gaps is the same distributed-control challenge that swarm robotics has been working on since its beginnings (■ahin, 2005; Rubenstein, Cornejo and Nagpal, 2014).

The learning system carries its own risks, and they deserve stating plainly. A swarm that shares what it learns can also share what it learns wrongly — a mistaken repair strategy could propagate through the population as readily as a good one, which argues for conservative bounds on what the swarm may attempt autonomously, validation before a model update is adopted fleet-wide, and the ability for operators to roll a swarm back to a known-good model. Autonomy of this kind should be earned incrementally, which is exactly why the staged concept begins with inspection alone.

7. Pathway to Reality

Each ingredient exists in laboratory form. Swarm robotics has demonstrated coordinated collective behaviour at genuine scale — a thousand simple robots self-assembling into shapes with no central control (Rubenstein, Cornejo and Nagpal, 2014) — and the field's foundations are two decades old (■ahin, 2005). Insect-scale robots that climb and cling using dry adhesives and micro-mechanisms are an active research area. Federated learning is deployed at scale on Earth today (McMahan et al., 2017). And free-flying and crawling robotic inspectors have already operated aboard crewed stations, establishing the operational precedent for machines that share a spacecraft with people. The credible near-term step is the inspection swarm — a genuinely useful capability that requires no autonomous repair and no unproven judgement — with intervention added only as the swarm's demonstrated competence earns it.

8. Conclusion

A termite mound stands for decades not because it was built perfectly but because something is always repairing it. Spacecraft have never had that, and the vehicles that will carry people to other worlds cannot do without it. The Myrmidon Swarm proposes the humblest possible robots doing the least dramatic possible work — finding a crack the width of a hair and sealing it, ten thousand times, for thirty years — joined by a shared mind that ensures no lesson is ever learned twice. It is a machine built on the insight that durability is not a property of materials but a process of constant care, and that the ships which last longest will be the ones that are never, for a moment, left unattended.

References

Harvard referencing style (Cite Them Right).

Christiansen, E.L. (2003) *Meteoroid/Debris Shielding*. NASA Technical Paper TP-2003-210788. Houston, TX: NASA Johnson Space Center.

McMahan, B., Moore, E., Ramage, D., Hampson, S. and Arcas, B.A. (2017) 'Communication-efficient learning of deep networks from decentralized data', *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 54, pp. 1273-1282.

Rubenstein, M., Cornejo, A. and Nagpal, R. (2014) 'Programmable self-assembly in a thousand-robot swarm', *Science*, 345(6198), pp. 795-799.

Sahin, E. (2005) 'Swarm robotics: from sources of inspiration to domains of application', in Sahin, E. and Spears, W.M. (eds.) *Swarm Robotics: SAB 2004 International Workshop*. Berlin: Springer, pp. 10-20.